The US-based FAA and cargo airlines such as UPS and FedEx are already testing the feasibility of GPS-based alternatives, which can help increase safety and efficiencies in controlling aircraft arrivals and departures. Using new digital tools, pilots would be able to get precise information from their cockpit displays about their position, the positions of other planes and maps of a particular airport and its runways. This would be enabled by a combination of satellite positioning technology and digital datalinks between aircraft.

If Boeing and the others can convince the sceptics that this system has to be adopted, it will become necessary for each and every one of the world's 14,000-plus commercial jets, military and private aircraft to have a costly receiver installed. The likelihood of this happening in the immediate future seems remote, but over the long run satellites may be the only way to enable aircraft to fly more safely, ease congestion, and prevent flight delays that seem to be an everyday occurrence.

The new radio

Most of the applications for satellites have traditionally been centred around communication, whether you consider the extremely successful services for consumers like television broadcasting, moderately successful ones like satellite-based Internet access, or ones that just did not take off—like the Iridium telephone communications network.

One new service that has begun to take off all over the world is satellite radio. Radio has been an extremely popular medium since before World War II. Even in the age of television and the Internet, radio continues to be popular amongst a wide range of audiences. Traditionally, the problem areas for radio have been the quality and range of the broadcast. Despite the popularity of thousands of stations that have adopted the Internet as a medium for streaming music to listeners all over the world, these issues have not been resolved completely.

Satellite radio promises to deliver a solution to both of these problems. It is probably the first significant advancement in radio since the arrival of FM more than 30 years ago. It offers listeners access to their favourite programmes while on the move, when they are out of range of conventional radio stations. Picture this: static free CD quality music from your favorite station, beamed to you in your car, home, or personal stereo—anywhere in the world. This is going to be possible because digital technology is actually being incorporated into the radio receiver set itself.

Sirrus, XM Radio and Worldcast are the pioneers that are looking to bring satellite radio to consumers all over the world. All of them will probably use proprietary chipsets in their receivers that will allow them to decipher the encrypted broadcast signal. The signal itself will be transmitted via a network of two or three satellites and several ground-based repeating stations.

These services are targeted primarily at car radio listeners, who will have access to a large selection of channels. The receivers are already being offered by car manufacturers in the US, including Ford, BMW and DiamlerChrysler, as options on their newer models. Users will have to pay a monthly subscription charge for the service, which will include many commercial-free channels. However, several of the larger manufacturers of consumer electronics like Hitachi, Panasonic and JVC have introduced digital receivers for the home device markets.

Future scope

Although a large number of people are already dependent on satellites, mostly for communication, these 'eyes in the sky' are now going to make an impact on our everyday lives in a more significant manner. What kind of help will these satellites render unto humankind in the future? Even Nostradamus would have been hard-pressed to answer this question. But what is certain is that we shall continue to become increasingly dependent on these sentinels that orbit our planet.

Car Watcher

Beware, all car thieves!

US-based Seaguard Electronics and Directed Electronics have developed Valet Car Com II, a system designed to provide a Web-enabled theft tracking system that can be accessed from any PC connected to the Internet. So if your Maruti gets stolen, just use this cyber 'valet' to track down the missing car.

WorldSpace becoming a Radio Star?

In India, US-based broadcaster, WorldSpace, first test-launched its radio services in Bangalore in September 2000. The company claims that the response has been quite positive. WorldSpace is offering several models of satellite radio receivers, ranging from Rs 4,400 to Rs 9,990.

They have signed up with Radio Mid-Day and AIR, in addition to other popular international broadcasters. WorldSpace also creates some original programmes, including modern rock and contemporary pop hits as well as broadcasts for children and adults.

Music, education and entertainment programmes in a multitude of languages that include English, Hindi, Tamil and Telugu are beamed by the AsiaStar satellite to subscribers across Asia.